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Communication-Avoiding Krylov Subspace Methods for Option Pricing

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Write acknowledgements to your supervisor, classmates, friends, family, partner... anyone who supported you during the MSc.

ABSTRACT

A trading desk repricing a multi-asset option book revalues the same partial differential equation thousands of times a day, and at production resolution a single price costs on the order of hundreds of seconds. The bottleneck is not arithmetic: a Krylov exponential integrator spends over 70% of its time in sparse matrix–vector products that move far more bytes than they compute, and most of the rest in an orthogonalization whose every step is a global synchronization. Communication-avoiding (CA) reformulations attack exactly these two costs, but whether the trade pays depends on the operator, the grid, and the machine.

This thesis answers *when* it pays, in coordinates that transfer. A benchmark pricer implementing five methods on two 3-asset European options establishes an uncompromised baseline and reproduces the reference results of Niesen and Wright. A regime study then builds a dimensionless map of the CA design space: a vertical axis R_v (working set over aggregate last-level cache) governing matrix-powers tiling, and a horizontal axis R_h (reduction cost over compute between reductions) governing s -step orthogonalization, with predicted crossovers at 1. The map’s coordinates are predictors computed before any run; measurements test the boundaries and never enter a coordinate. The certified s -step block width is predicted from the spectrum alone and verified integer-exact on 20 of 21 runs; a log-robustness law bounds the damage non-normality does to that prediction; and a closed-form tensor law gives the growth of eigenvector conditioning with asset dimension. On the measurement side the study delivers one null result with a mechanism (the horizontal reduction cut does not move wall-clock, exactly as the map predicts), one shortfall with a mechanism (cache-blocked matrix powers trades halo arithmetic for DRAM traffic at a loss), and a placed trajectory showing the production operator to be squarely in vertical-only territory. A planned GPU study, outlined here, will test the map’s central portability claim on a memory system that differs in kind.

Keywords: Krylov subspace methods, communication-avoiding algorithms, matrix exponential, option pricing, roofline model, s -step methods, performance modeling

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1 INTRODUCTION

A trading desk repricing a multi-asset book revalues the same PDE thousands of times a day, so the wall-clock cost of one solve sets what can be quoted, hedged, and risk-managed intraday. For a 3-asset basket at a resolution fine enough to matter, the published cost is roughly 700s per price [15]. The bottleneck is not arithmetic. A Krylov exponential integrator spends over 70% of its time in sparse matrix–vector products (SpMV) that move far more bytes than they compute, and the rest in an orthogonalization whose every step is a global synchronization. Both are *communication* costs: data crossing the memory hierarchy, and cores waiting on each other [2, 10]. Communication-avoiding (CA) reformulations attack exactly these—matrix powers computes s Krylov vectors per pass over the operator instead of one [6, 14], and s -step orthogonalization replaces $O(m^2)$ reductions with $O(m/s)$ [10, 5]—trading redundant arithmetic, which is cheap, for movement and latency, which are not.

Whether that trade pays is not a property of the algorithm. It depends on the operator, the grid, and the machine. This thesis answers *when* it pays, and does so in coordinates that transfer.

Two artifacts support that answer. A **benchmark pricer** in C++23 implements five methods (Crank–Nicolson, two ADI variants, a one-shot matrix exponential, and a Krylov exponential integrator) on two 3-asset European options, and establishes the uncompromised baseline the CA work must beat (Section 5). A **regime study** (Section 8) builds a dimensionless map of the CA design space: because both of its axes are ratios rather than absolute times, the map is independent of the machine and of the problem, so a reader can place their own operator and hardware on it and read off which mechanism, if either, can pay for them. The CA Krylov exponential integrator itself is not yet implemented; the map exists to decide what to build.

The design rule that keeps the map non-circular is stated up front because everything else depends on it: *coordinates are predictors*, computed before a run from the problem size, the sparsity pattern, the spectrum, and the machine constants; *outcomes* (wall-clock, orthogonality loss, achieved bytes) test the boundaries and never enter a coordinate.

The thesis proceeds as follows. Section 2 states the pricing problem and its discretization. Section 3 presents the Krylov exponential integrator, its cost structure, and the CA reformulations that target it. Section 4 describes the experimental platform and measurement methodology. Section 5 reproduces the Niesen–Wright benchmark as a control. Section 6 profiles the solver and places its kernels on a cache-aware roofline. Section 7 presents the OpenMP scaling study that motivates the two CA mechanisms. Section 8 develops the regime map and its numerical and measured results. Section 9 outlines the GPU study now in progress, which is designed as a falsification test of the map’s central claim. Section 10 concludes.

2 THE PRICING PROBLEM AND ITS DISCRETIZATION

The model problem is the 3-asset Black–Scholes PDE [4] in log-price coordinates, solved backward in pseudo-time τ from 0 (payoff) to T (price):

$$\frac{\partial u}{\partial \tau} = \frac{1}{2} \sum_{d,d'} \rho_{dd'} \sigma_d \sigma_{d'} \frac{\partial^2 u}{\partial x_d \partial x_{d'}} + \sum_d \left(r - \frac{1}{2} \sigma_d^2 \right) \frac{\partial u}{\partial x_d} - r u + b(\tau). \quad (2.0.1)$$

Two payoffs are priced. The basket call,

$$u_0(x) = \max \left(\sum_d w_d e^{x_d} - K, 0 \right),$$

and the rainbow min-call,

$$u_0(x) = \max\left(\min_d e^{x_d} - K, 0\right).$$

The boundary forcing term $b(\tau)$ for the basket case is encoded as a polynomial $B \cdot s(\tau)$ with $s(\tau) = [\tau^2/2, \tau, 1]^\top$, derived from a deep in-the-money approximation on ghost nodes outside the grid. The rainbow case uses modified finite-difference stencils (a zero-gamma boundary condition) instead.

Discretization is by second-order central differences on a uniform Cartesian grid with x_1 -fastest lexicographic ordering, so every operator is a three-fold Kronecker product with identity in the undifferentiated directions. The assembled $\mathbf{A}$ carries at most 19 nonzeros per row (a 7-point stencil plus 3×4 mixed-derivative diagonals). Its bandwidth is $O(N_1 N_2)$, set by the x_3 and mixed terms, which rules out banded direct solvers at scale. $\mathbf{A}$ splits as $\mathbf{A}_{\text{sym}} + \mathbf{A}_{\text{skew}}$ —diffusion, mixed, and reaction terms symmetric, convection skew-symmetric—so it is not symmetric in general, but $\text{Re}(\lambda) \leq 0$ holds, which is what stability of $e^{h\mathbf{A}}$ requires. Table 1 lists the default model parameters used throughout.

Table 1: Default model parameters.

Parameter	Value	Description
Spot prices S_0	(100, 100, 100)	Initial asset prices
Strike K	100	Option strike
Risk-free rate r	0.04	Continuously compounded
Maturity T	1.0 year	Fixed
Volatilities σ	(0.30, 0.35, 0.40)	Per-asset annual volatility
Correlations ρ	(0.50, 0.50, 0.50)	Off-diagonal pairs $(\rho_{01}, \rho_{02}, \rho_{12})$
Basket weights w	(1/3, 1/3, 1/3)	Equal-weight basket
Grid half-width α	2.85	Domain $\pm\alpha\sigma\sqrt{T}$ per dimension

Stiffness grows sharply with refinement: the smallest eigenvalue moves from -24 at grid size $n = 11$ to -2924 at $n = 101$ (where $N = n^3$ is the total number of unknowns). Over-solving in time on a coarse grid is therefore pointless, and the commercial motivation sits at the other end: Niesen and Wright report ~ 700 s for Hundsdorfer–Verwer to reach one basis point at 61^3 [15].

3 THE KRYLOV EXPONENTIAL INTEGRATOR

After spatial discretization, the semi-discrete system takes the affine form

$$\frac{d\mathbf{u}}{d\tau} = \mathbf{A}\mathbf{u} + \mathbf{b}, \quad \mathbf{u}(\tau_0) = \mathbf{u}_0. \quad (3.0.1)$$

Following Niesen and Wright [15], the inhomogeneity is absorbed into an *extended system*: define

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0}^T & 0 \end{bmatrix} \in \mathbb{R}^{(N+1) \times (N+1)}, \quad \tilde{\mathbf{v}} = \begin{bmatrix} \mathbf{u}_0 \\ 1 \end{bmatrix} \in \mathbb{R}^{N+1},$$

so that the exact solution at time $\tau_0 + h$ is encoded in the first N components of $e^{h\tilde{\mathbf{A}}}\tilde{\mathbf{v}}$.

The Krylov approximation proceeds via the Arnoldi iteration on $\tilde{\mathbf{A}}$ with starting vector $\tilde{\mathbf{v}}$, producing an orthonormal basis $\mathbf{V}_m$ and an upper Hessenberg matrix $\bar{\mathbf{H}}_m \in \mathbb{R}^{(m+1) \times m}$ [16, 9]. Dropping the last row gives $\mathbf{H}_m \in \mathbb{R}^{m \times m}$, and the approximation reads

$$e^{h\tilde{\mathbf{A}}}\tilde{\mathbf{v}} \approx \beta \mathbf{V}_m e^{h\mathbf{H}_m} \mathbf{e}_1, \quad \beta = \|\tilde{\mathbf{v}}\|. \quad (3.0.2)$$

Algorithm 1 gives the full adaptive procedure, and Algorithm 2 the Arnoldi iteration it calls.

Algorithm 1: Adaptive Krylov Exponential Integrator for $\mathbf{u}' = \mathbf{A}\mathbf{u} + \mathbf{b}$

Input: Matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$, vector $\mathbf{b} \in \mathbb{R}^N$, initial condition $\mathbf{u}_0$, final time T , time-step $\Delta\tau$, tolerance tol

Output: Solution snapshots $\{\mathbf{u}_k\}_{k=0}^K$ at $\tau_k = k \Delta\tau$

```

1  $\tau \leftarrow 0$ ;
2 for  $k = 0, 1, \dots, K - 1$  do
3    $h \leftarrow \min(\Delta\tau, T - \tau)$ ;
4    $m \leftarrow 1$ ;  $\text{err} \leftarrow \infty$ ;

   /* Form extended system */
5    $\tilde{\mathbf{A}} \leftarrow \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0}^T & 0 \end{bmatrix} \in \mathbb{R}^{(N+1) \times (N+1)}$ ;
6    $\tilde{\mathbf{v}} \leftarrow \begin{bmatrix} \mathbf{u}_k \\ 1 \end{bmatrix} \in \mathbb{R}^{N+1}$ ;

   /* Adaptive Krylov loop */
7   while  $m \leq N + 1$  do
8     Run  $m$  steps of Arnoldi on  $(\tilde{\mathbf{A}}, \tilde{\mathbf{v}}) \Rightarrow \mathbf{V}_{m+1}, \bar{\mathbf{H}}_m$  // Algorithm 2
9     Store  $\bar{h}_{m+1,m} \leftarrow [\bar{\mathbf{H}}_m]_{m+1,m}$  // sub-diagonal entry
10     $\mathbf{H}_m \leftarrow \bar{\mathbf{H}}_m(1:m, 1:m)$  // drop last row
11     $\beta \leftarrow \|\tilde{\mathbf{v}}\|$ ;
12     $\mathbf{f} \leftarrow e^{h\mathbf{H}_m} \mathbf{e}_1$  // small  $m \times m$  matrix exponential
13     $\mathbf{w} \leftarrow \beta \mathbf{V}_m \mathbf{f}$  // Krylov approximation in full space

    /* A posteriori error estimate */
14     $\text{err} \leftarrow |\bar{h}_{m+1,m}| \cdot \|\mathbf{f}\|_m$ ;
15    if  $\text{err} < \text{tol}$  then
16      break;
17    end
18     $m \leftarrow m + 1$ ;
19  end
20   $\mathbf{u}_{k+1} \leftarrow \mathbf{w}(1:N)$  // extract first  $N$  components
21   $\tau \leftarrow \tau + h$ ;
22 end

```

Algorithm 2: Arnoldi Iteration

Input: Matrix $\tilde{\mathbf{A}} \in \mathbb{R}^{(N+1) \times (N+1)}$, vector $\tilde{\mathbf{v}} \in \mathbb{R}^{N+1}$, subspace dimension m , breakdown tolerance `tol`

Output: Orthonormal basis $\mathbf{V}_{m+1} \in \mathbb{R}^{(N+1) \times (m+1)}$, upper Hessenberg $\bar{\mathbf{H}}_m \in \mathbb{R}^{(m+1) \times m}$

```

1  $\beta \leftarrow \|\tilde{\mathbf{v}}\|$ ;  $\mathbf{v}_1 \leftarrow \tilde{\mathbf{v}}/\beta$ ;
2 for  $j = 1, \dots, m$  do
3    $\mathbf{w} \leftarrow \tilde{\mathbf{A}} \mathbf{v}_j$ ;
4   for  $i = 1, \dots, j$  do
5     // Modified Gram-Schmidt
6      $h_{i,j} \leftarrow \mathbf{v}_i^T \mathbf{w}$ ;
7      $\mathbf{w} \leftarrow \mathbf{w} - h_{i,j} \mathbf{v}_i$ ;
8   end
9    $h_{j+1,j} \leftarrow \|\mathbf{w}\|$ ;
10  if  $h_{j+1,j} < \text{tol}$  then
11    break // Invariant subspace found (happy breakdown)
12  end
13   $\mathbf{v}_{j+1} \leftarrow \mathbf{w}/h_{j+1,j}$ ;
14 end
```

3.1 COST STRUCTURE AND THE SEQUENTIAL DEPENDENCY

The dominant cost of the integrator is the Arnoldi iteration. Each step j requires one sparse matrix–vector product $\mathbf{w} = \tilde{\mathbf{A}}\mathbf{v}_j$, costing $O(N)$, and orthogonalization against all previous vectors, costing $O(jN)$. Crucially, step j cannot begin until $\mathbf{v}_j$ is available, which requires step $j-1$ to have completed. This creates a strictly sequential data dependency across the Krylov dimension m . The individual SpMV and inner products within each step can be parallelized, but this inner parallelism scales poorly: the SpMV is bandwidth-bound, and every inner product in the modified Gram–Schmidt (MGS) loop is a global reduction. These are the two *communication* walls, and they are different in kind:

Table 2: The two kernels, their bounds, and the CA mechanism each motivates.

Kernel	Bound	Communication	CA mechanism
SpMV	bandwidth (memory hierarchy)	vertical	matrix powers / cache blocking
Gram–Schmidt	synchronization (reductions)	horizontal	s -step / block orthogonalization

3.2 COMMUNICATION-AVOIDING REFORMULATIONS

The s -step (CA-Arnoldi) reformulation [10, 6, 5] computes s basis vectors $\mathbf{v}, \mathbf{A}\mathbf{v}, \dots, \mathbf{A}^s\mathbf{v}$ per communication phase. The *matrix-powers kernel* [14] computes all s products in one pass over the operator, tiling the grid so that each panel and its s -deep halo stay cache-resident—attacking the vertical wall. The block is then orthogonalized at once—here by CholeskyQR2 [19], whose certificate $\kappa(\mathbf{B}) \leq u^{-1/2}$ (with u the unit roundoff) is a floating-point theorem rather than a machine property—attacking the horizontal wall with $O(m/s)$ reductions in place of MGS’s $O(m^2)$. The cost is conditioning: the monomial basis $[\mathbf{v}, \mathbf{A}\mathbf{v}, \dots, \mathbf{A}^s\mathbf{v}]$ grows geometrically ill-conditioned in s , which caps the usable block width. Quantifying that cap—predicting it from the spectrum, certifying it, and measuring where the prediction survives non-normality—is a central thread of Section 8.

An alternative not pursued here is the rational Krylov subspace [3], which replaces the sequential polynomial recurrence with m independent shifted solves. Its parallelism is attractive, but each solve is itself a sparse linear system whose iterative solution re-exports both communication walls, and the near-optimal pole selection adds problem-specific tuning. The s -step route was chosen because its costs are analyzable in machine terms—which is precisely what the regime map requires.

4 EXPERIMENTAL PLATFORM AND METHODOLOGY

Every result in this thesis was gathered on a single bare-metal workstation (“puffin”): an AMD Ryzen Threadripper 3960X, 24 cores in a single socket and a single NUMA node, organized as 8 core complexes (CCX) of 3 cores, each CCX with its own 16 MiB slice of L3 (128 MiB total). Per-core caches are 32 KiB L1d and 512 KiB L2; nominal maximum clock is 3.8 GHz, and main memory is 4-channel DDR4-3200 with a theoretical 102.4 GB/s roof. The machine also hosts an NVIDIA RTX 3090 (24 GB), which plays a role in the GPU outlook of Section 9. The machine’s derived constants live in a single header (`include/machine.hpp`) under the key `amd-3960x`, and every predicted coordinate in the regime study is computed from them.

WHY ONE BARE-METAL MACHINE, AND NOT A CLOUD VM. An earlier cloud arm was dropped, on two grounds. Contention: timing variance that looks internally consistent is simultaneously evidence of memory-bound behavior and of noise from co-tenants, and from inside the guest the two are not separable. Topology, the sharper objection: the hypervisor reports a socket and a private last-level cache slice *per vCPU*—a fiction of the virtualization rather than the silicon. Both coordinates of the regime map are functions of real cache geometry (one is a cache-capacity ratio, the other is set by how many cache-domain boundaries a reduction tree crosses), so neither is knowable on an invented topology, and the map cannot be drawn there at all. Bare metal removes the hypervisor but not OS scheduling noise, so every timed quantity below is reported as a median with an inter-quartile range over repeated trials; on a boost-enabled chip timing noise is right-skewed, and the median reports a *sustainable* throughput where the minimum would report an unsustainable boost spike.

VECTORIZATION IS NOT THE BASELINE’S PROBLEM. A STREAM Triad [13] sweep built with default flags (SSE2) and with `-march=native` (AVX2) is identical inside run-to-run spread—22.8 vs. 22.6 GB/s at one thread, 39.3 vs. 39.5 GB/s at the 24-thread saturation knee, against the 102.4 GB/s theoretical roof (Figure 1). On a bandwidth-bound kernel the instruction set buys nothing, so the CPU baseline is fair as measured.

REPRODUCIBILITY. Every number quoted in this thesis names the script that reproduces it. The repository builds with CMake and GCC 16 (C++23); sweep scripts write CSVs under `data/`, which is generated locally and excluded from version control precisely so that no quoted figure can drift from its generating run; each figure is produced by a self-contained plotting script with inline dependency metadata. The regime executables read machine constants from a single audited header, so porting the study to a new machine is a matter of deriving one new constants block—a property the GPU study of Section 9 depends on.

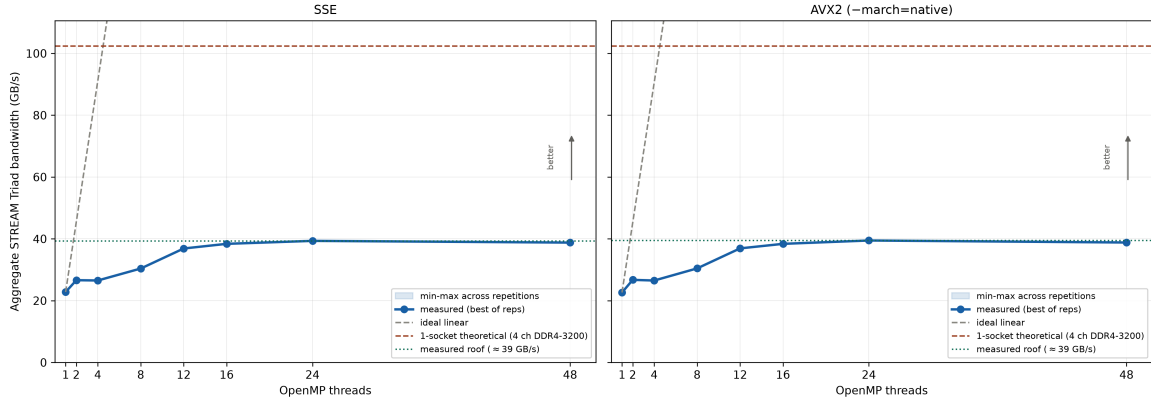


Figure 1: STREAM Triad bandwidth sweep, SSE2 vs. AVX2 builds. The two instruction sets are indistinguishable at every thread count: the kernel is bandwidth-bound, not vectorization-bound.

5 BENCHMARK: REPRODUCING NIESEN–WRIGHT

Five methods are compared. Three are time-steppers over a fixed number of substeps: Crank–Nicolson ($\theta = 0.5$, fully implicit, second order, one sparse LU factorization per run), and two alternating-direction-implicit splittings—Douglas–Rachford [7], with three direction-split solves per step, and Hundsdorfer–Verwer [11, 12], a Douglas–Rachford predictor with a corrector. The remaining two exponentiate the operator: **ME** forms $e^{T\tilde{\mathbf{A}}}$ in one shot by scaling and squaring [8, 1] with an ∞ -norm early exit, choosing its own substep count; **KSM-EI** is the Krylov exponential integrator of Section 3, growing the basis until an a-posteriori estimate meets the tolerance.

The benchmark reproduces Figure 1 of Niesen and Wright [15]: log-log plots of ODE error against CPU time for all five methods, across both option types and both grid sizes (Figure 2).

THE REFEREE. The measurement is against the exact ODE solution, not the analytic price. Before each option type is benchmarked, a high-accuracy matrix-exponential referee is computed at maximum Taylor degree $m = 55$ and $\theta = 9.9$, giving $s = \lceil T\|\tilde{\mathbf{A}}\|_1/\theta \rceil$ internal substeps at a tolerance of $2^{-53} \approx 1.1 \times 10^{-16}$. Scoring against the analytic price instead would fold spatial discretization error into a claim about time integration, so the referee is deliberately a different object from the contestants: the exact solution of the same semi-discrete system. The error is the Euclidean norm over the $9 \times 9 \times 9$ grid neighborhood centered on the spot rather than the value at a single node, so a lucky cancellation at one grid point cannot flatter a method.

CONTROL, NOT RESULT. The method ordering matches theory, and that is the point of this section: it is the *control* for everything after it. On an autonomous linear problem the one-shot matrix exponential is genuinely hard to beat: at $n = 15$ time-exactness buys nothing, because every method is already pinned at the spatial error (0.066). The Krylov integrator’s structural advantages—adaptivity, no step-count trial-and-error, and a retained trajectory that a one-shot exponential discards—only cash out away from this benign regime, at the production resolutions where the ~ 700 s cost lives.

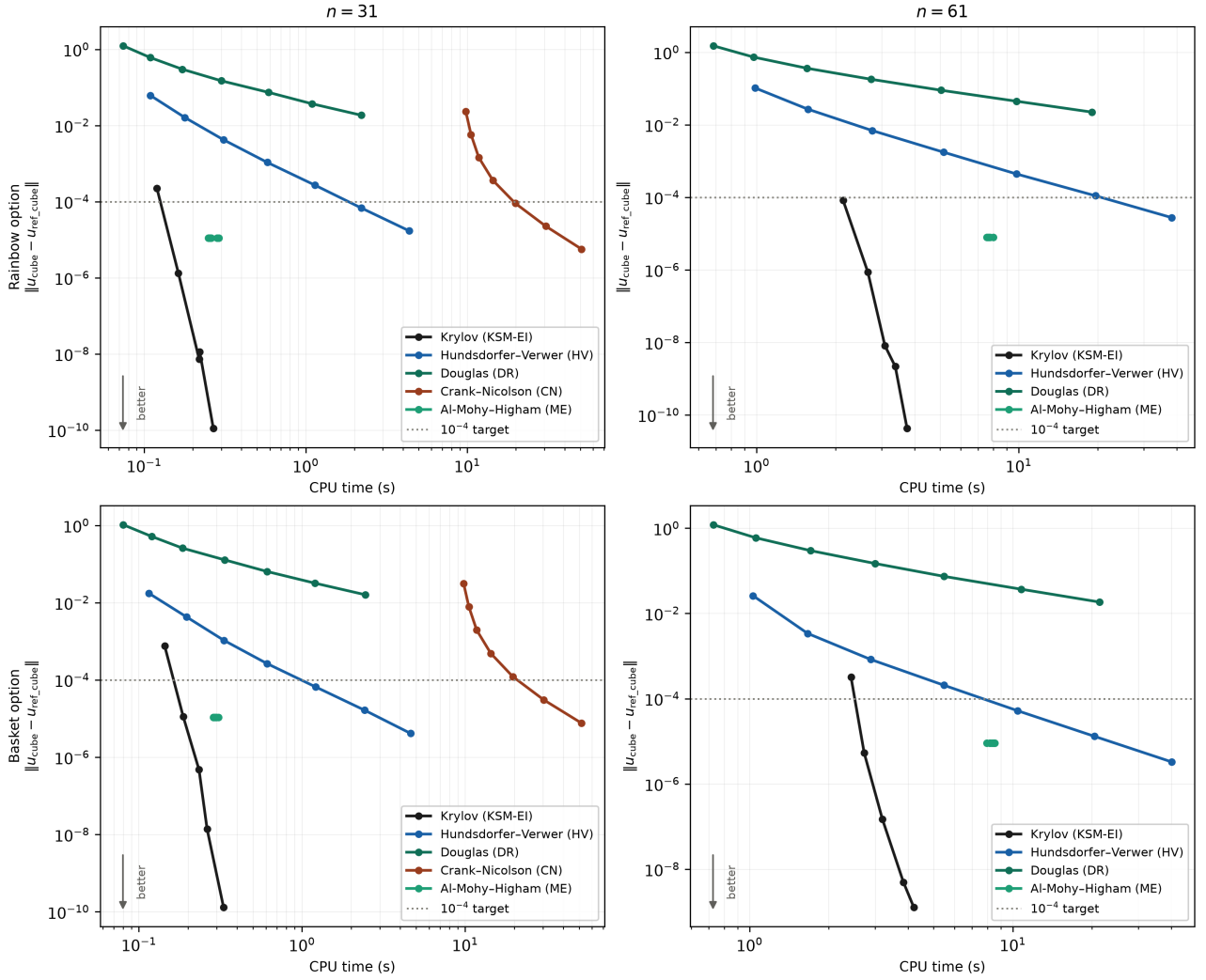


Figure 2: ODE error vs. CPU time for the five methods, across both payoffs and grid sizes, reproducing Niesen–Wright Figure 1. The Krylov integrator dominates in the $\sim 10^{-4}$ accuracy band; Crank–Nicolson’s non-monotone stiff-mode ringing ($R(-\infty) = -1$) is visible as predicted.

6 PROFILE AND ROOFLINE

At $n = 61$ (226,981 unknowns, 4.23M nonzeros, ≈ 49 MB in compressed sparse column form) over 100 steps, the KSM-EI solve is overwhelmingly SpMV and Gram–Schmidt; the dense $\exp(\mathbf{H}_m)$ is noise (Table 3). Gram–Schmidt’s share grows with the grid because the Krylov dimension m grows and MGS is quadratic in m .

Table 3: Kernel shares of KSM-EI solve time (basket, tolerance 10^{-8}). $\bar{m}$ is the mean Krylov dimension.

Grid	$\bar{m}$	SpMV	Gram–Schmidt	dense expm	other
$n = 31$	7.60	82.9%	11.7%	0.40%	4.9%
$n = 53$	8.42	77.7%	16.8%	0.34%	5.1%
$n = 65$	9.24	76.0%	18.0%	0.21%	5.9%
$n = 73$	9.89	75.2%	18.8%	0.15%	5.9%
$n = 89$	11.34	70.3%	23.8%	0.08%	5.8%

Both kernels sit far below the compute ceiling. The cache-aware roofline [18] shows why, and where the CA opportunity is (Figure 3, Table 4). Single-core roofs on puffin, all measured: DRAM 20.9 GB/s, L3 58.2 GB/s, L2 88.6 GB/s, compute peak 68.1 GFLOP/s (an FMA loop).

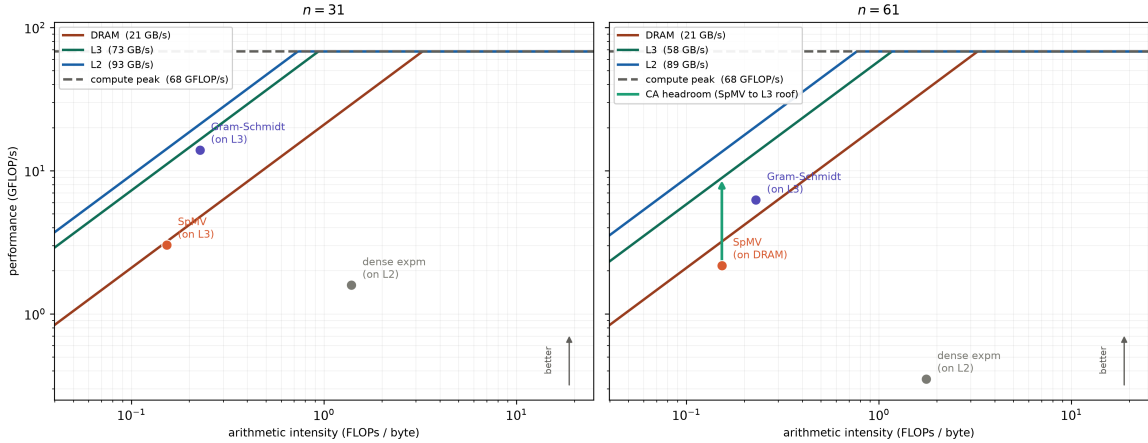


Figure 3: Cache-aware roofline for the two hot kernels at $n = 31$ and $n = 61$. At $n = 61$ the SpMV has spilled to DRAM; the vertical distance from the measured point to the L3 roof at the same arithmetic intensity is exactly the traffic a matrix-powers reformulation would recover.

Table 4: Measured kernel positions on the roofline. AI is arithmetic intensity.

Grid	Kernel	AI (FLOP/B)	Achieved (GFLOP/s)	Resident tier
$n = 31$	SpMV	0.153	3.02	L3
$n = 31$	Gram–Schmidt	0.227	13.91	L3
$n = 61$	SpMV	0.153	2.17	DRAM
$n = 61$	Gram–Schmidt	0.230	6.24	L3

The attainable gain from moving a kernel one tier up the hierarchy is

$$\text{gain} = \frac{\min(BW_{\text{fast}} \cdot AI, \text{peak})}{\min(BW_{\text{slow}} \cdot AI, \text{peak})}, \quad (6.0.1)$$

which collapses to the bandwidth ratio only while *both* roofs are still sloped. It must bend down once the fast roof reaches the compute ceiling—which the $s \times$ -higher arithmetic intensity of a matrix-powers kernel can actually reach. Quoting a flat bandwidth ratio as “the CA speedup” overstates the ceiling.

7 THE OPENMP SCALING STUDY

The scaling study measures how the two hot kernels of one Arnoldi cycle respond to added cores—it does not measure solution accuracy. The thesis of the experiment is that the two kernels fail to scale for two *different* reasons, each motivating a distinct CA mechanism (Table 2); this split is what the regime analysis of Section 8 later turns into two independent dimensionless axes.

7.1 INSTRUMENT DESIGN

To make both ceilings visible and attributable, the harness freezes the Krylov dimension at $m = 8$ with the convergence check disabled, so the Gram–Schmidt/SpMV work ratio does not drift and any runtime change across the core count P is attributable to hardware rather than a changing algorithm (the genuine $m(n)$ growth is measured separately). It uses a hand-rolled CSR, row-partitioned SpMV in which each thread is the sole writer of a contiguous, disjoint slice of the output vector, with partition seams aligned to the cache line; it runs MGS with parallel global reductions, exposing the synchronization cost directly; and it reports per-kernel timings so a single sweep yields SpMV’s plateau and Gram–Schmidt’s roll-off on one axis. Runs pin threads with `OMP_PLACES=cores` and `OMP_PROC_BIND=close`, so each CCX fills before the next engages and per-CCX L3 bandwidth steps appear as discrete risers; the swept core counts are CCX-boundary aligned and SMT lanes are excluded. Each point is the median of five to seven timed repeats after a warm-up, with the inter-quartile range as the error bar.

Two initialization arms form a controlled A/B experiment on memory placement: arm A fills the matrix from thread 0 (first-touch parks it on one memory placement), arm B has each thread first-touch exactly the row partition it will later compute on. The two arms turn out to be indistinguishable—for a reason that is itself the finding, taken up in Section 7.3.

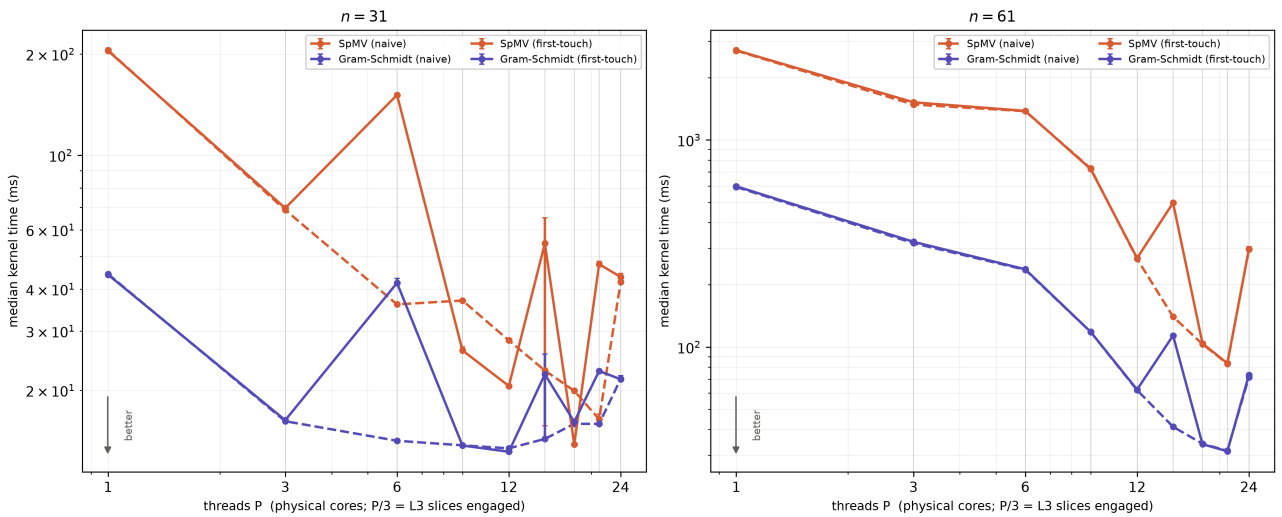


Figure 4: Per-kernel strong scaling at $n = 31$ and $n = 61$, both initialization arms. SpMV shows a plateau at $n = 31$ and a superlinear transition at $n = 61$; Gram–Schmidt shows the synchronization staircase.

7.2 THE QUARANTINED $m(n)$ GROWTH

The harness freezes m so that the scaling curves measure hardware; the genuine growth of the Krylov dimension with the grid is measured separately and reported as an independent numerical-analysis finding (Figure 6). The mean $\bar{m}$ per step grows from 7.6 at $n = 31$ to 11.3 at $n = 89$ —slow growth,

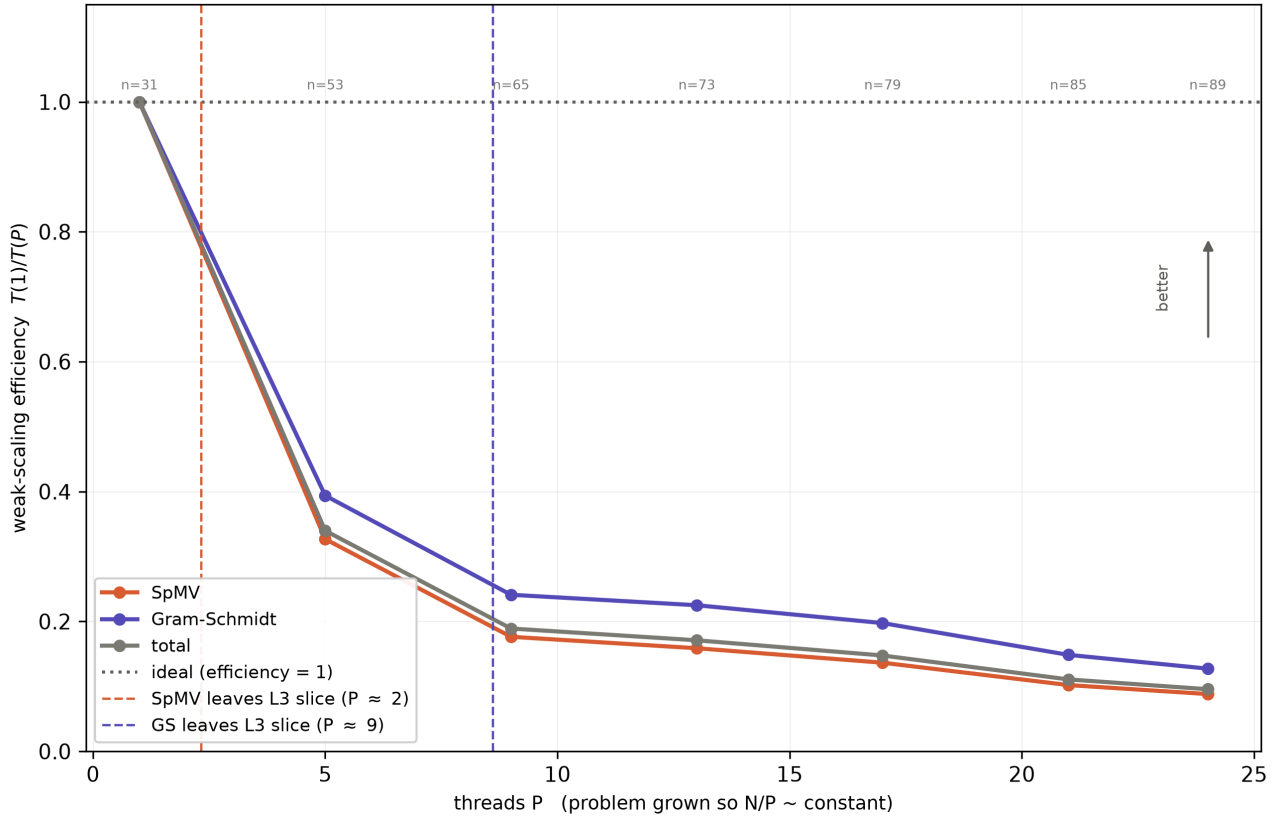


Figure 5: Per-kernel weak-scaling efficiency $T(1)/T(P)$ with $n(P) = n_1 P^{1/3}$ rounded odd, cache-tier crossings annotated.

but enough to drive Gram–Schmidt’s quadratic share upward (Table 3), and the input to the small- m safety argument of Section 8.4.

7.3 WHY THE TRANSITION HAPPENS: THE LOCALITY CONTROL

At $n = 61$ the SpMV runtime drops *superlinearly* ($36.7\times$ on 24 cores) with a knee at $P = 12$ (Figure 4). The hypothesis: once each CCX’s share of the matrix ($3 \times 52.8/P$ MiB) fits in its own 16 MiB L3 slice and stays resident across the repeated SpMVs, reads come from L3 instead of DRAM. Quantitatively this checks out—the implied aggregate bandwidth crosses the socket DRAM ceiling (~ 95 GB/s) exactly at $P = 12$, which is impossible from DRAM and therefore proves L3 residency.

The initialization arms *cannot* test this: the 3960X is single-NUMA, and Zen 2 L3 is a per-CCX victim cache, so L3 residency is set by which core *accesses* a row, not by which thread first-touched the page. Both arms use the same row partition and produce identical curves. The real controlled experiment is over the *compute schedule*, which changes only which rows each core reads (Table 5, Figure 7).

The refuted *cyclic* prediction is a result in its own right, and it sharpens the CA argument. Per-CCX *volume* is not sufficient for residency: dealing cache-line blocks round-robin leaves each CCX holding the same number of bytes scattered across the whole matrix, and the arm collapses to roughly the *rotate* curve ($7.0\times$ vs. $6.6\times$) rather than the *block* one ($36.7\times$). **Contiguity, not volume, is what keeps a slice resident**—which is exactly the property a matrix-powers tiling is built to exploit, and exactly the property the scattered arm of the vertical sweep (Section 8.9) destroys on purpose. The identical arm-A/arm-B result thereby stops being an embarrassment and becomes the control that establishes the transition as a locality effect—the “before” that the CA matrix-powers reformulation

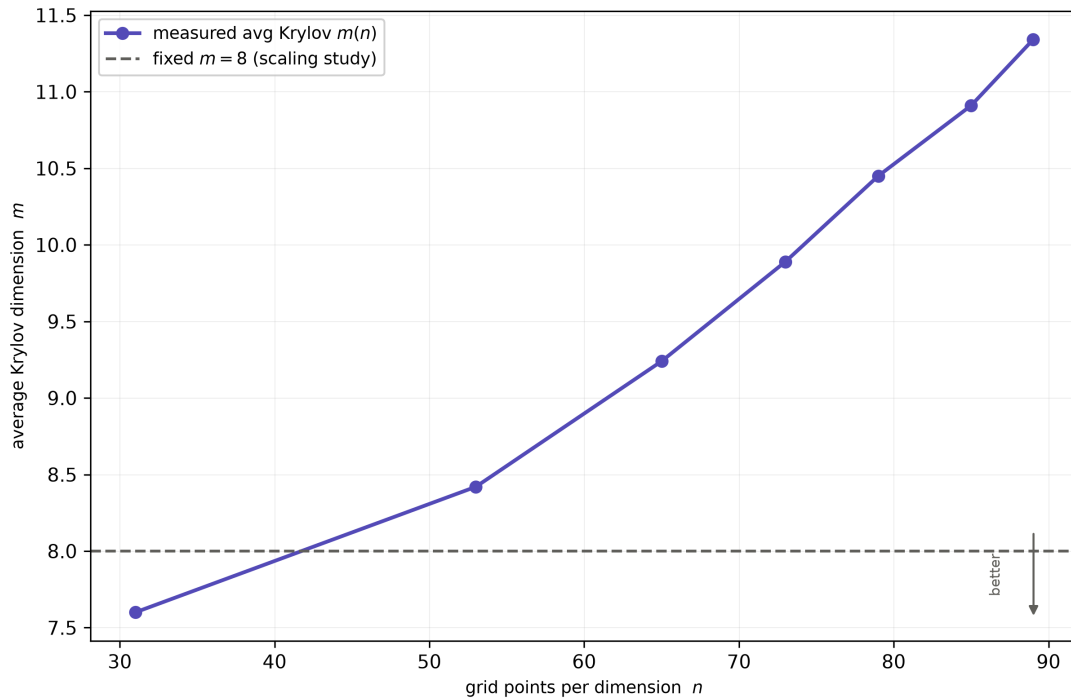


Figure 6: The $m(n)$ Krylov-growth curve, measured with the convergence check enabled and quarantined from the scaling instrument, which pins $m = 8$.

Table 5: The locality control at $n = 61$: three compute schedules, their predictions, and the measurements (SpMV speedup at $P = 24$; implied aggregate bandwidth at $P = 12$).

Schedule	What it does	Prediction	Measured
block	contiguous band per core, fixed across SpMVs	transition at $P = 12$ (locality-preserving control)	confirmed: $36.7\times$, knee at $P = 12$, 176 GB/s
cyclic	cache-line blocks dealt round-robin, fixed	still transitions—scatter leaves per-CCX volume unchanged	refuted: $7.0\times$, no knee, 36 GB/s
rotate	contiguous bands, rotating by one each SpMV	transition vanishes—no CCX retains a cache-fitting subset	confirmed: $6.6\times$, no knee, 66 GB/s

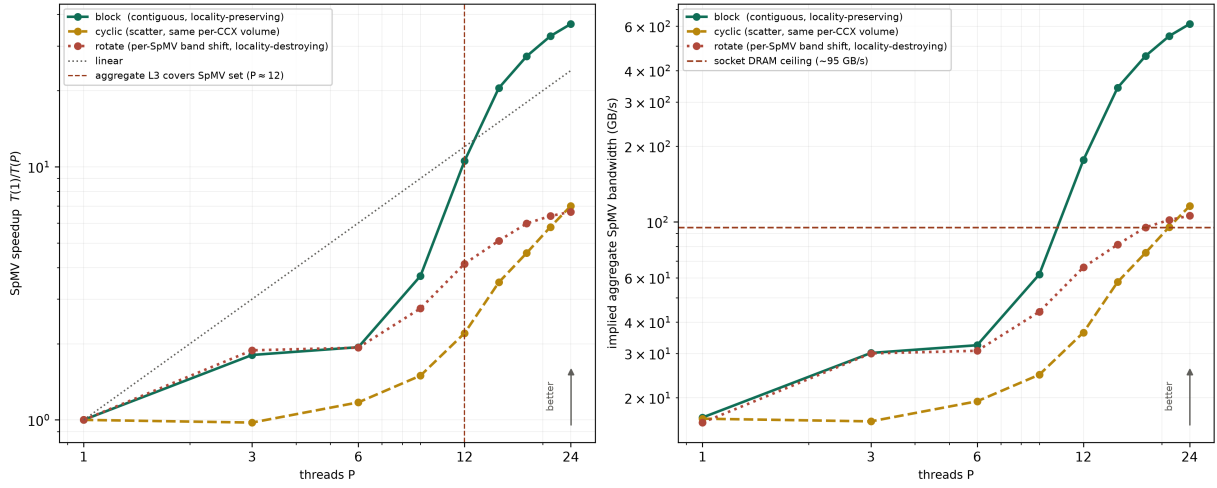


Figure 7: The locality control. Only **block** carries an aggregate bandwidth DRAM cannot supply, so the transition is genuinely L3 residency, not a scheduling coincidence.

improves on.

8 REGIME ANALYSIS: A DIMENSIONLESS MAP

The scaling study says the baseline solver is memory-bound and synchronization-bound. It does not say whether a CA reformulation would *help*, or where. The regime analysis answers that with a dimensionless map whose two axes are the two mechanisms, with a predicted crossover at 1 on each, plus a synthetic instrument that can be dialed to any point on the map and a real Black–Scholes operator that is *placed* on it by measurement.

The non-circularity rule of Section 1 is enforced structurally: the predictor lives in a header that never sees a timer (`regime.hpp`). In particular, R_h ’s numerator is a machine constant times a tree depth, never a measured fraction of runtime.

8.1 THE COORDINATES

Table 6: The two dimensionless coordinates.

Axis	Definition	Mechanism it governs	Scales like
R_v	Arnoldi working set / aggregate last-level cache	vertical: matrix-powers tiling, avoiding DRAM traffic	$\sim 1/P$
R_h	reduction cost / compute per core between reductions	horizontal: s -step, cutting the reduction count	$\sim P \log P$

Both crossovers are predicted at $R = 1$ ($\theta_v = \theta_h = 1$), giving four corners: neither mechanism (lower-left), horizontal only (lower-right), vertical only (upper-left), and both (upper-right). The Krylov dimension m was demoted from an axis to a *modulator*: it moves working sets, so it moves R_v , but it is not itself a portable coordinate—two machines with the same m can sit on opposite sides of θ_v .

R_h ’s numerator needs a magnitude, not just a shape in P , or θ_h cannot be drawn at all. A calibration instrument measures it directly as a bare scalar all-reduce over an empty team—no payload, no

application—and the model fitted to it is

$$t_{\text{reduce}}(P) = t_{\text{intra}} \cdot \text{intra levels}(P) + t_{\text{cross}} \cdot \min(\text{cross levels}(P), L^*). \quad (8.1.1)$$

On puffin: $t_{\text{intra}} = 120.8 \text{ ns}$, $t_{\text{cross}} = 652.8 \text{ ns}$ (a cross-CCX tree level costs $5.4 \times$ an intra-CCX one), and $L^* = 2.20$ —past roughly two cache-domain crossings, subtree concurrency hides the rest. Measured tree all-reduce latency runs from 9.6 ns at $P = 1$ to $1.70 \mu\text{s}$ at $P = 24$ ($R^2 = 0.98$ against the model). The cost is set by how many cache-domain boundaries the tree crosses, not by core count—which is precisely why this coordinate cannot be measured on a virtualized topology (Section 4).

8.2 THE INSTRUMENT AND ITS GATES

The numerics gate is not a timing run: nothing is threaded or timed, and no result depends on the host. It establishes that the synthetic instrument’s a-priori arithmetic predictions hold before any communication claim is built on them (Table 7).

Table 7: The controls. A gate failure is fatal to everything downstream; a finding is a result in its own right.

Control	Claim under test	Status
C1	the Kronecker-sum spectrum is analytic and correct	gate
C2	the scatter knob is an exact similarity transform (so a pure- R_v dial exists)	gate
C3	measured m obeys the Hochbruck–Lubich spectral bound [9]	gate (Hermitian PSD)
C3b	a spectral shift leaves m fixed	finding
C4	κ of the matrix-powers basis equals the row-scaled Vandermonde from the spectrum	gate when normal
C5	the CholeskyQR certificate $\kappa \leq u^{-1/2}$ [19] is sound	gate
C5b	worst-case block count over the physical ensemble (“small- m safety”)	finding
C6	where the operator is non-normal, does the spectral prediction survive?	finding

C2 is load-bearing. The scatter knob permutes the operator symmetrically, $\mathbf{PAP}^\top$, which moves bytes and provably nothing else; if it moved the spectrum, the “pure R_v line” would be an illusion and every vertical result would be confounded. The sweep runs in five phases—gate, confound, m -range, non-normal, and finally the real discretized Black–Scholes operator with its actual discounted payoff, swept across grids. That last phase is *transfer by measurement*: it lands the real operator on the same (R_v, R_h) plane as the synthetic instrument.

8.3 BASIS CONDITIONING AND THE CONFOUND CONTROL

The certified s -step block width s_{max} is predicted from the spectrum alone (a row-scaled Vandermonde [17]), then measured. Where a closed-form spectrum exists the prediction is **integer-exact on 20 of 21 runs**; the single exception is the most violently non-normal point in the sweep ($\gamma = 0.4$, $\kappa(X) = 7.7 \times 10^8$) and is off by one step. Every deviation larger than that lives in the variable-advection arm—the family the log-robustness law (Section 8.5) explicitly excludes. The real Black–Scholes operator deviates by at most one step at any grid measured (Figure 8).

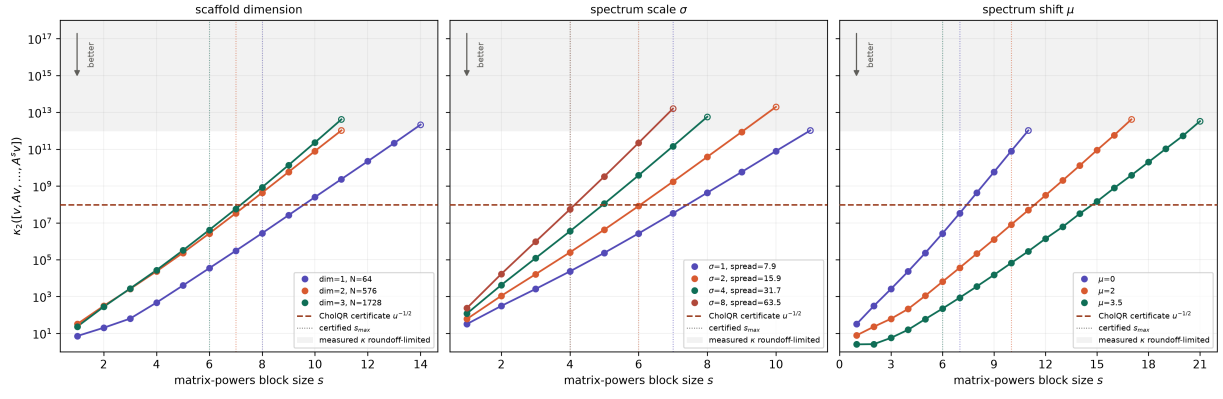


Figure 8: Predicted vs. measured basis condition number and the certified block width $s_{\max}$ across the swept operator family.

The obvious confound is that $s_{\max}$ might just be tracking the Krylov dimension m . It does not (Figure 9): with the spectrum pinned and m moved through 5–12 (via the step size and tolerance), $s_{\max}$ stays flat at 7; with m pinned at 5 and the spectrum scaled and shifted, $s_{\max}$ ranges from 4 to 14. Safety is a property of the spectrum, not of the Krylov dimension.

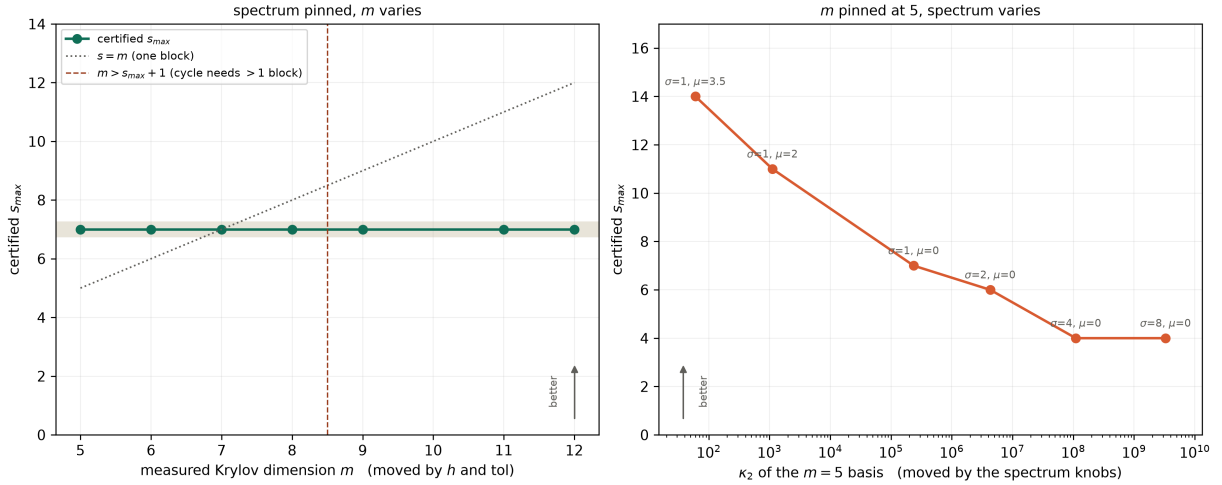


Figure 9: The confound control: $s_{\max}$ under pinned spectrum with varying m (flat), and pinned m with varying spectrum (4–14).

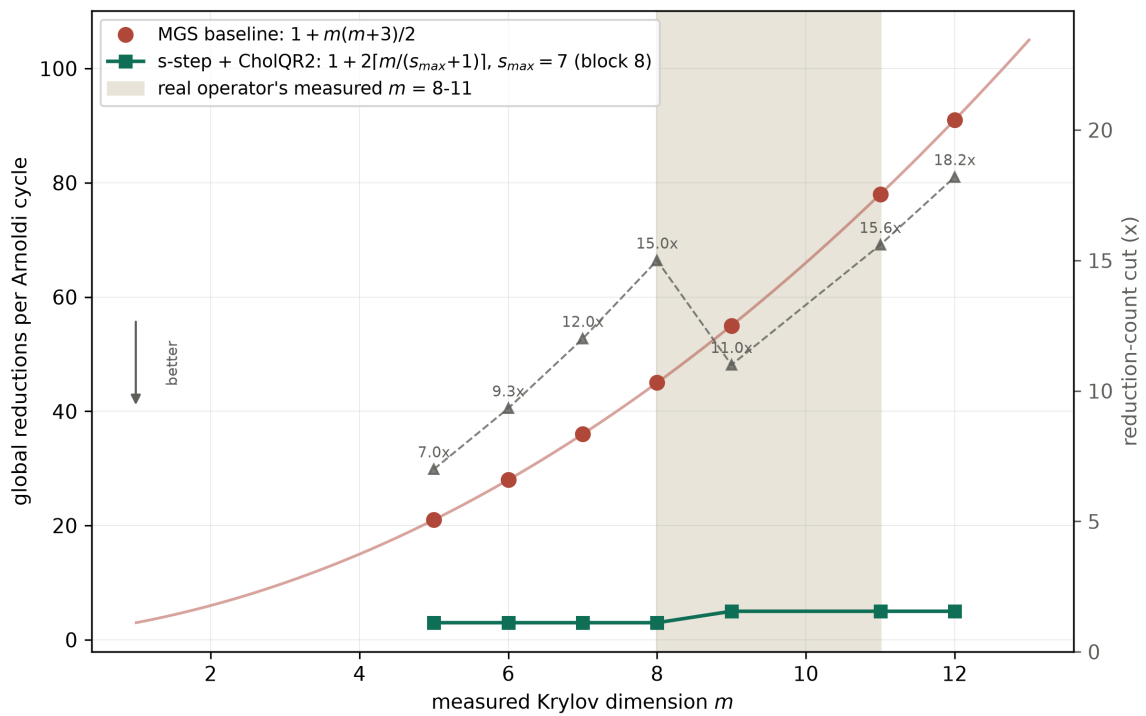
8.4 THE REDUCTION LEDGER AND SMALL- m SAFETY

MGS performs $1 + m(m + 3)/2$ global reductions per Arnoldi cycle. The stable s -step arm (matrix powers + CholeskyQR2) performs $1 + 2\lceil m/(s_{\max} + 1) \rceil$ (Table 8). Across the real problem's measured window $m = 8$ –11 that is an **11–16 $\times$ cut in reduction count** at the $s_{\max} = 7$ certified for this spectrum (Figure 10).

SMALL- m SAFETY. The real operator's Krylov dimension is small and grows slowly: $m = 6$ –8 across the eigensolvable grids $n = 10$ –20, and 7.6 to 11.3 across $n = 31$ –89 in production. Because a certified block yields $s_{\max} + 1$ vectors, an entire Arnoldi cycle fits in **2–3 blocks**. This is the argument that makes s -step viable for this integrator at all: the machinery never has to run deep enough for monomial-basis conditioning to become the dominant risk.

Table 8: The reduction ledger at $s_{\max} = 7$.

m	MGS reductions	s -step + CholQR2	cut
5	21	3	$7.0\times$
8	45	3	$15.0\times$
9	55	5	$11.0\times$
11	78	5	$15.6\times$
12	91	5	$18.2\times$

Figure 10: Reductions per Arnoldi cycle: MGS vs. the certified s -step arm.

8.5 NON-NORMALITY: THE LOG-ROBUSTNESS LAW

The spectral prediction of $s_{\max}$ assumes normality. The real operator is not normal (mixed-derivative cross terms plus convection). The rigorous two-sided bound is that the prediction error in decades is at most $\log_{10} \kappa(X)$, the eigenvector conditioning. Measured, the data sits far below that line and grows only *logarithmically* (Figure 11): the fitted slope of (error in decades) against $\log_{10} \kappa(X)$ is **0.069** for the constant-coefficient mechanisms (advection, correlation, and their combination) and **0.250**—about $4\times$ steeper—for variable-coefficient advection (a position-dependent ramp). The two slopes differ by a factor of four, so the law is *scoped to the constant-coefficient family*, not mechanism-independent—which is exactly why the variable-advection knob exists in the instrument.

Both slopes are fitted on the canonical $\kappa(X)$ axis (Section 8.6). This matters for the constant-coefficient arm, whose zero-correlation half is a pure Kronecker sum: its axes are interchangeable, so the dense $\kappa(X)$ there is degenerate and inflated by up to $173\times$. Fitted on the dense axis the slope reads 0.061 instead of 0.069—small enough not to disturb the conclusion, but the two arms would have been compared on axes of different validity.

A *decade* here is one factor of ten in the basis condition number; the unit is chosen because it converts directly into block width. Each extra power appended to $[\mathbf{v}, \mathbf{A}\mathbf{v}, \dots, \mathbf{A}^s\mathbf{v}]$ multiplies its condition number by a roughly constant factor, measured at a median of 1.05 decades per power over the swept family (inter-quartile range 0.95 to 1.12), so the figure marks one s -step at 1.2 decades. An error below that line cannot move the certified block width by a full step.

The real Black–Scholes operator is placed on this figure as a test point, excluded from every fit. Its prediction error runs 0.19 to 0.85 decades at $\kappa(X)$ up to 2.3×10^3 , comfortably **under one s -step** at every grid measured.

Operator by operator, what the spectral prediction needs is normality; definiteness and separability are conveniences (Table 9).

Table 9: Which structural properties the spectral prediction of $s_{\max}$ actually needs.

Operator	Normal?	Definite?	Separable?	Spectral prediction
Laplacian	yes	PSD	yes	exact (closed form)
shift μ	yes	indefinite	yes	exact ($\kappa(X) = 1$)
correlation ρ	yes	PSD	no	exact (dense eigensolve)
constant γ advection	no	—	yes	survives (< 1 step)
$\rho + \gamma$	no	—	no	survives (< 1 step)
variable γ advection	no	—	no	must be measured (> 1 step)

8.6 THE ASSET-DIMENSION LAW

How does $\kappa(X)$ grow with the number of assets d ? This axis matters because the application grows the asset count, not the grid: a refinement study swept the grid at fixed $d = 3$ and found $\kappa(X)$ flat, which confirms the mesh-Péclet reading ($\log \kappa(X) \sim d n_1 \gamma$ with $\gamma \sim C/n_1$, so n_1 cancels) and says nothing about d .

The law is linear in d , with a slope known in closed form. The synthetic operator applies its cross term to exactly one axis pair at every d , so it factors as a Kronecker sum $\mathbf{A}_d = \mathbf{B}_{01} \oplus \mathbf{T} \oplus \dots \oplus \mathbf{T}$. A Kronecker product multiplies singular values, so

$$\kappa(X) = \kappa(X_{01}) \kappa(X_1)^{d-2} \quad \implies \quad \log_{10} \kappa(X) = 0.40499 d + 0.01422, \quad (8.6.1)$$

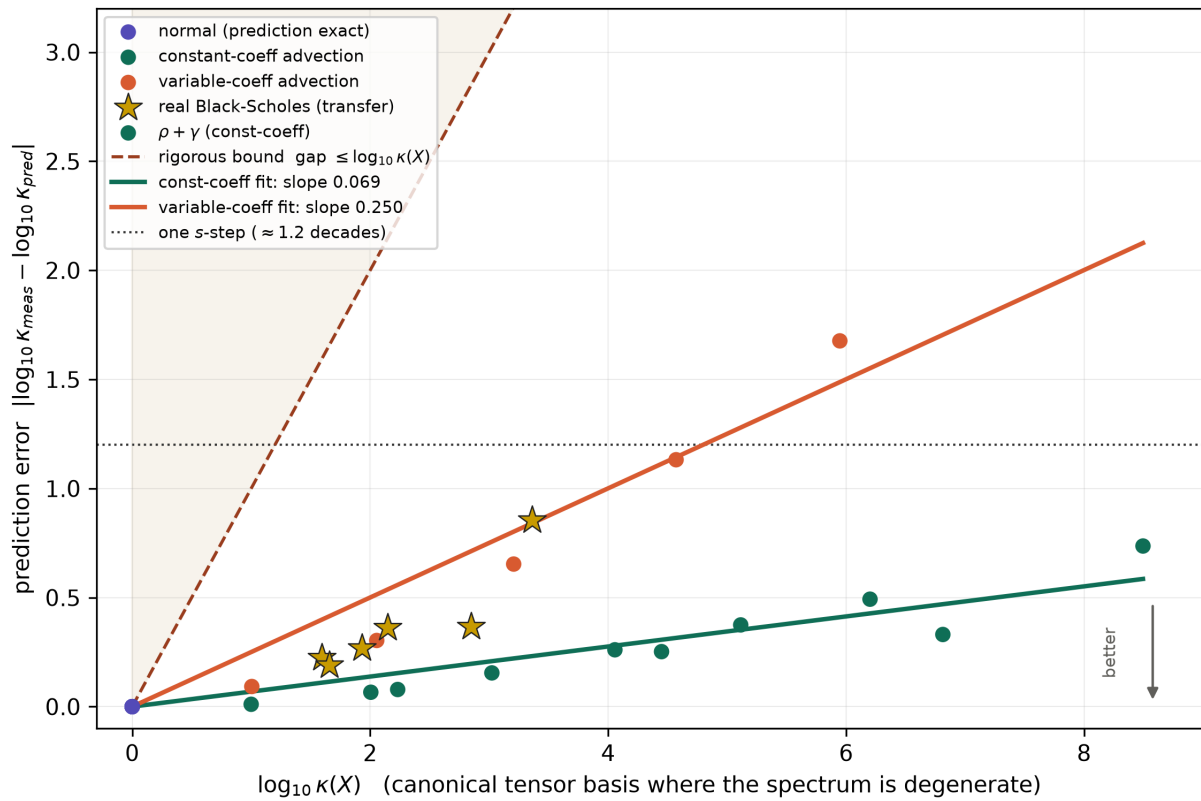


Figure 11: The log-robustness law: prediction error in decades vs. $\log_{10} \kappa(X)$. The rigorous bound is the identity line; the measured data grows logarithmically, with the constant-coefficient and variable-coefficient families fitted separately. The real Black–Scholes operator (test point, excluded from fits) sits under one s -step at every grid.

with the slope $0.404995 = \log_{10} \kappa(X_1)$ analytic. The observed $d = 2 \rightarrow 3$ increment is 0.40500, and the formula reproduces both degeneracy-free points to 3×10^{-7} .

The dense $\kappa(X)$ that a straightforward eigensolve returns must *not* be fitted along d , because from $d = 4$ it is not a property of the operator at all. Axes $2, \dots, d-1$ carry no cross term, so interchanging any two of them is an exact symmetry of $\mathbf{A}$: the spectrum repeats, and inside a repeated eigenspace every basis is admissible, so the value reported is whichever basis the eigensolver’s rounding happened to produce. A symmetric permutation $\mathbf{PAP}^\top$ is an exact similarity, so any function of the operator is invariant under it—yet the dense value moves by a factor of 3.84 at $d = 4$ and 7.76 at $d = 5$ across eight permutations once the spectrum is degenerate. The study therefore reports a canonical, basis-independent value (`kappa_X_struct`, computed per Kronecker factor) alongside the dense one (Table 10, Figure 12).

Table 10: Canonical vs. dense $\kappa(X)$ along the asset dimension.

d	N	canonical $\kappa(X)$	dense $\kappa(X)$	spectrum
2	16	6.671	6.671	simple (agree to 3×10^{-7})
3	64	16.95	16.95	simple (agree to 3×10^{-7})
4	256	43.07	109.5	degenerate: dense meaningless
5	1024	109.4	5.2×10^3	degenerate: dense meaningless
6	4096	278.1	3.7×10^4	degenerate: dense meaningless

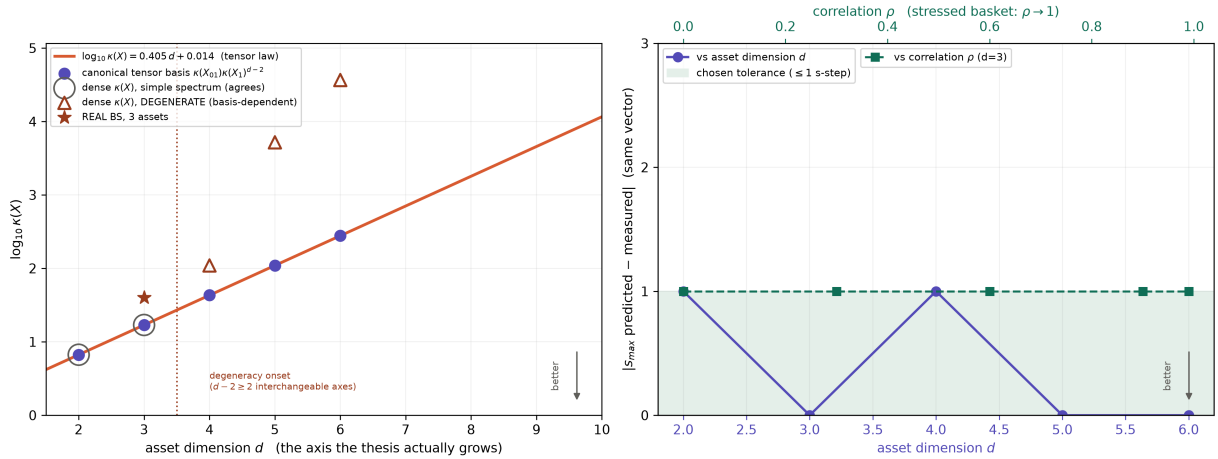


Figure 12: The asset-dimension law: canonical $\kappa(X)$ grows geometrically in d with an analytic slope; the dense value diverges from it exactly where the spectrum degenerates.

The correlation arm ($n_1 = 8$, $d = 3$) is sound wherever the cross term is actually present, and its $\rho = 0$ endpoint is the most degenerate point in the study (only 89 of 512 eigenvalues distinct). With the canonical value as anchor (Table 11), **correlation raises $\kappa(X)$ monotonically**, from 688 at $\rho = 0$ to 3405 at $\rho = 0.99$ —about $5\times$ across the full range.

Table 11: The correlation arm ($n_1 = 8$, $d = 3$). At $\rho > 0$ the dense value matches the canonical one exactly; at $\rho = 0$ the operator is a pure Kronecker sum and the dense value is degenerate.

ρ	0	0.3	0.6	0.9	0.99
canonical $\kappa(X)$	688.0	851.8	1566.9	2277.9	3405.2
dense $\kappa(X)$	<i>11350</i>	851.8	1566.9	2277.9	3405.2

An earlier reading had correlation *reducing* $\kappa(X)$, but that took the degenerate $\rho = 0$ dense value (1.14×10^4) as its anchor, and permuting that same operator returns anything from 4×10^4 to 1.5×10^6 :

there was no fall to explain. Dimension still dominates— ρ across its entire admissible range buys about 1.7 assets' worth of conditioning, and $\rho \leq 1$ while d is unbounded—but correlation never helps.

The practical consequence: **measure the block width at high d rather than trusting a conservative a-priori bound**, because $\kappa(X)$ grows geometrically in d while staying flat under grid refinement. The law's scope is the synthetic instrument: the real basket operator couples all three asset pairs, so the tensor identity, which rests on one coupled pair, does not extend to it.

8.7 THE COMPUTABILITY GAP

The certificate is validated by dense eigensolve, which caps the control at $N \leq 32,768$, while the performance claims live at $N \sim 10^5$. The proposed bridge was to exploit the constant-coefficient Toeplitz structure for an analytic spectrum, avoiding the eigensolve entirely. **That route is closed.** The cross term factors exactly as $c(\mathbf{D} \otimes \mathbf{D})$ with $\mathbf{D} = \text{tridiag}(-1, 0, +1)$ (verified to machine zero). A diagonal similarity $\text{diag}(z^j)$ symmetrizes $\text{tridiag}(b, a, c)$ iff $z^2 = c/b$ —and the two terms demand incompatible z on the same axis: $z^2 = (1 - \gamma)/(1 + \gamma)$ for advection–diffusion against $z^2 = -1$ for the cross factor. Numerically, the sine basis leaves 3.7% off-diagonal mass with the cross term against 2.8×10^{-14} without, and no single z helps. The obstruction persists at $\gamma = 0$, so it is the Dirichlet boundary condition, not the advection.

The consolation is constructive: the tensor identity (8.6.1) gives $\kappa(X)$ at any d from a single $n_1^2 \times n_1^2$ eigensolve—no $N \times N$ solve and no cap—for the synthetic operator. The gap remains open for the real Black–Scholes operator, whose all-pairs correlation breaks the Kronecker factorization.

8.8 THE HORIZONTAL MECHANISM: A MEASURED NULL RESULT

The reduction ledger promises an 11–16 $\times$ cut in reduction *count*. The wall-clock does not move. Two independent measurements say why, and they agree:

1. **The primitive was never the bottleneck.** Replacing the naive barrier-plus- $O(P)$ -scan reduction with a proper binary tree—a 1.7 $\times$ improvement in the bare all-reduce latency at $P = 24$ (2.91 μs down to 1.70 μs)—leaves Gram–Schmidt wall-clock unchanged to within 5% at every core count. The OpenMP runtime's barrier is already a tree, and the barrier itself dominates the scan.
2. **Gram–Schmidt is vector-bound, not reduction-bound.** MGS re-reads column $\mathbf{V}_i$ once for every $j \geq i$, so $m(m+1)/2$ column reads at $8N$ bytes each. That quadratic is a *bandwidth* term, and it is what the roll-off in the scaling study is actually made of.

The map predicted this before the experiment: at production resolution $n = 61$ the real operator sits at $R_h = 0.080 \ll \theta_h = 1$. The horizontal mechanism has no home on this machine, and the null result is a *confirmation* of the coordinate, not a failure of it. The thesis leads with the timed number, not the reduction count.

The upper-right corner—both mechanisms binding simultaneously—is structurally unreachable on puffin. It is a single-NUMA socket, so no tree in the reduction crosses anything more expensive than a CCX boundary; reaching θ_h at a working set that also exceeds θ_v needs a reduction cost roughly 25 $\times$ higher than any this machine can produce.

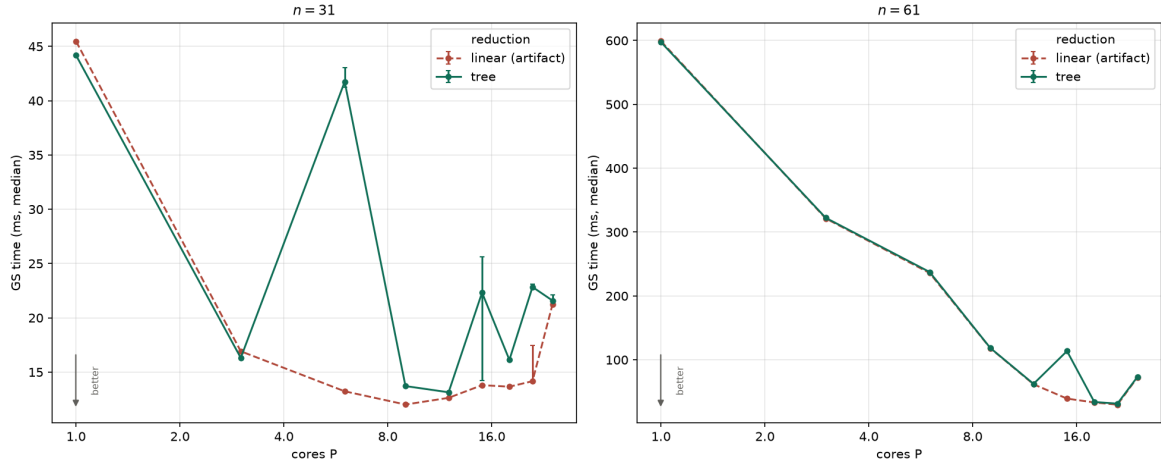


Figure 13: Gram-Schmidt wall-clock under the calibrated tree reduction vs. the $O(P)$ redundant scan it replaced ($n = 61$, arm B; one contaminated $P = 15$ launch, inflated across all kernels, is excluded). A $1.7\times$ faster primitive leaves the kernel unchanged within 5%.

8.9 THE VERTICAL MECHANISM: A MEASURED SHORTFALL

A timed sweep runs a cache-blocked matrix-powers kernel against the untiled baseline across grids that carry R_v from 0.035 to 86. The modeled traffic ratio—the ceiling the roofline says tiling could reach—is $4\text{--}8\times$. The measurement is not close (Table 12, Figure 14).

Table 12: The vertical sweep: measured speedup of the tiled matrix-powers kernel against the modeled traffic ratio.

Pattern	Tile level	R_v (1 core)	Measured speedup	Modeled ratio
banded	L3	0.035–2.69	0.72–1.31 $\times$	$4\text{--}8\times$
banded	L2	1.12–86.1	0.37–0.89 $\times$	$4\text{--}8\times$
scattered	L3	0.035–2.69	1.00 $\times$ (falls back to naive)	$4\times$

This is a **kernel ceiling, not a refuted boundary**: the tiled arm never reached a bandwidth-bound regime, so the experiment never got to test θ_v . A diagnostic kernel that skips the halo entirely (wrong basis, throughput only) separates the two tile levels cleanly (Figure 15). At the L2 level (small panels, fat halo), removing the halo recovers throughput toward baseline (1.17 rising to 2.13 against a 2.70 baseline at $n_1 = 400$): cache-blocking *works*, and the shortfall is redundant halo arithmetic traded for DRAM traffic—a bad trade at these panel sizes, not a broken kernel. At the L3 level (large panels, thin halo), removing the halo barely moves throughput (2.06 to 2.11 against 2.76): neither arm is bandwidth-bound, and both are capped by something insensitive to cache residency, most consistent with gather latency on the indirect column access.

The scattered arm delimits where the vertical mechanism cannot pay at all. Matrix-powers tiling is structurally impossible once the columns scatter, so the sweep falls back to the naive form and measures $1.00\times$ by construction. Since the scatter knob is a similarity transform (gate C2), that boundary is a property of the sparsity pattern alone, at fixed spectrum and fixed working set.

Two honest limitations. Nothing here is a *counted* DRAM measurement, because puffin’s uncore counters need root, so modeled bytes are reconciled against the roofline instead—weaker evidence, labeled as such on every figure. And the projected curve on Figure 14 is a model claim, drawn dashed.

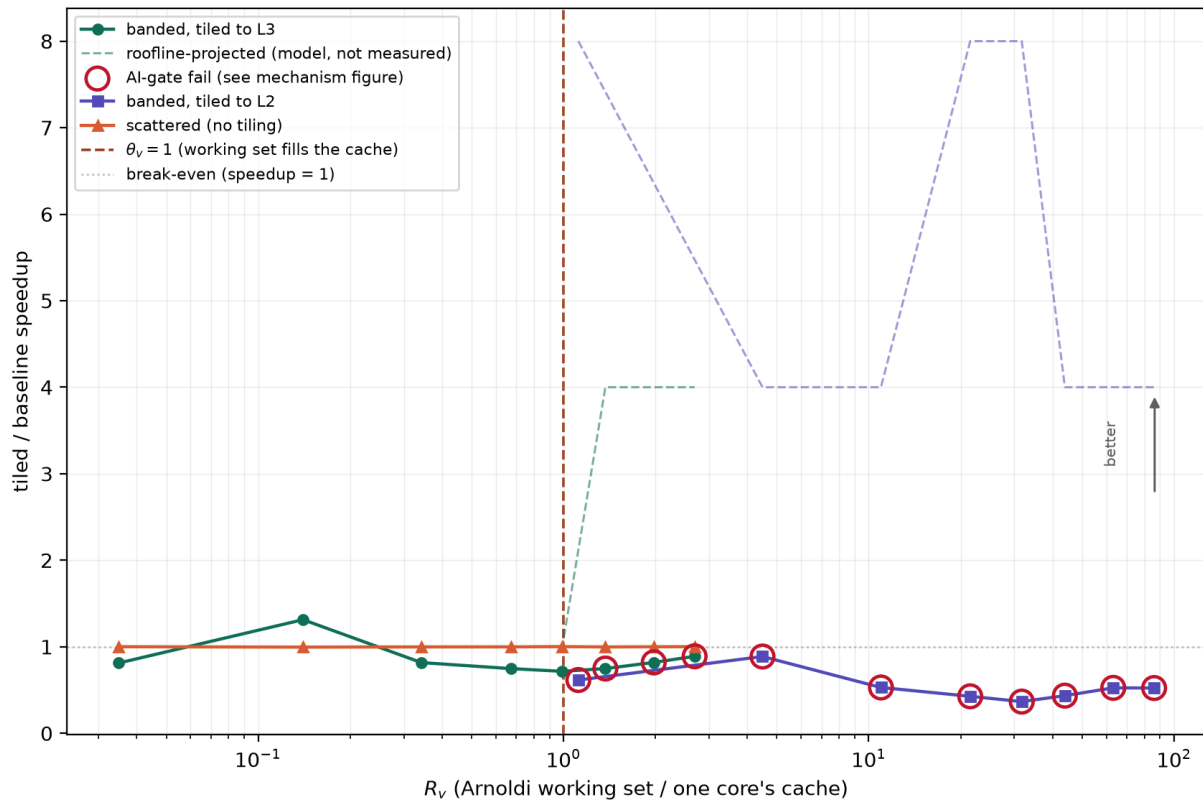


Figure 14: The vertical sweep: measured speedup of cache-blocked matrix powers across R_v . The projected curve is an explicit model claim, drawn dashed, never presented as data.

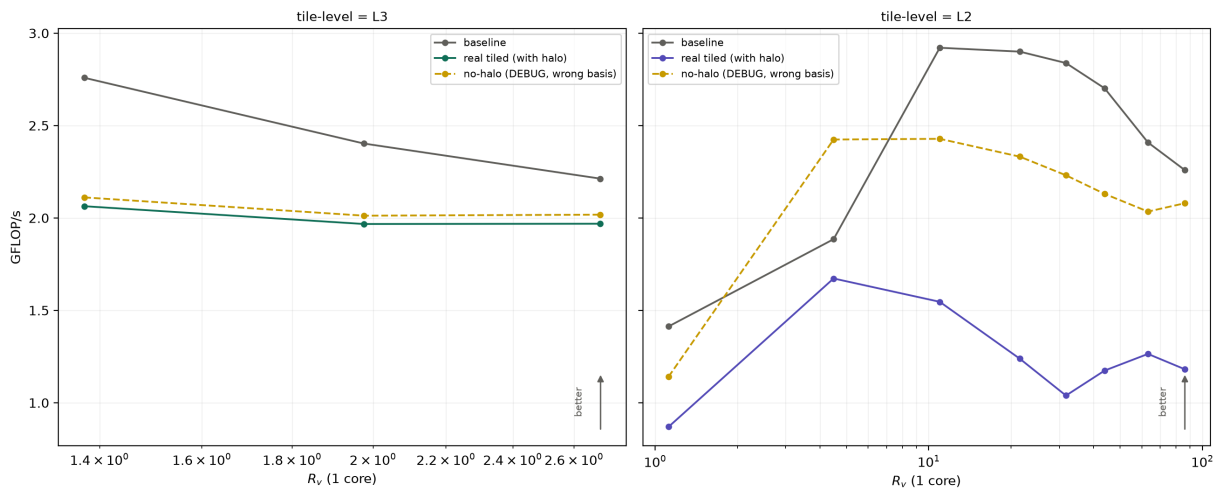


Figure 15: The no-halo diagnostic. L2 tiling recovers toward baseline without the halo (the halo trade is the shortfall); L3 tiling barely moves (the cap is insensitive to cache residency).

8.10 BLOCK WIDTH: TWO ROOFS, AND WHICH ONE BINDS

Everywhere else the block width s is pinned at the certified value, which hides a question. s is pushed *up* by both mechanisms—a wider block means fewer reductions and more powers per operator stream—and pulled *down* by two independent ceilings:

- **numerical (the certificate):** $\kappa([\mathbf{v}, \mathbf{A}\mathbf{v}, \dots, \mathbf{A}^s \mathbf{v}])$ grows geometrically in s ; past $u^{-1/2}$ the monomial basis leaves the CholeskyQR certificate, so the block builds but will not orthogonalize.
- **capacity (the ghost):** the halo is $s \cdot w$ rows per side, so a wider block shrinks the panel until none fits and the tiling collapses.

Which arrives first is a joint property of operator and machine, and on puffin it depends on the tiling level (Table 13, Figure 16).

Table 13: The two roofs on the block width. The L3 ghost roof is quoted as $\geq$ because a panel still fit at every swept s up to the sweep ceiling of 24; the certificate roof is measured directly ($\kappa(\mathbf{B}_s)$ crosses $u^{-1/2}$ between $s = 8$ and $s = 9$ at $n_1 = 420$).

Tile level	n_1	Certificate roof s	Ghost roof s	Binding roof
L3	420	8	≥ 24	numerical
L3	480, 560	6	≥ 24	numerical
L2	420	8	3	capacity
L2	480	6	3	capacity
L2	560	6	2	capacity

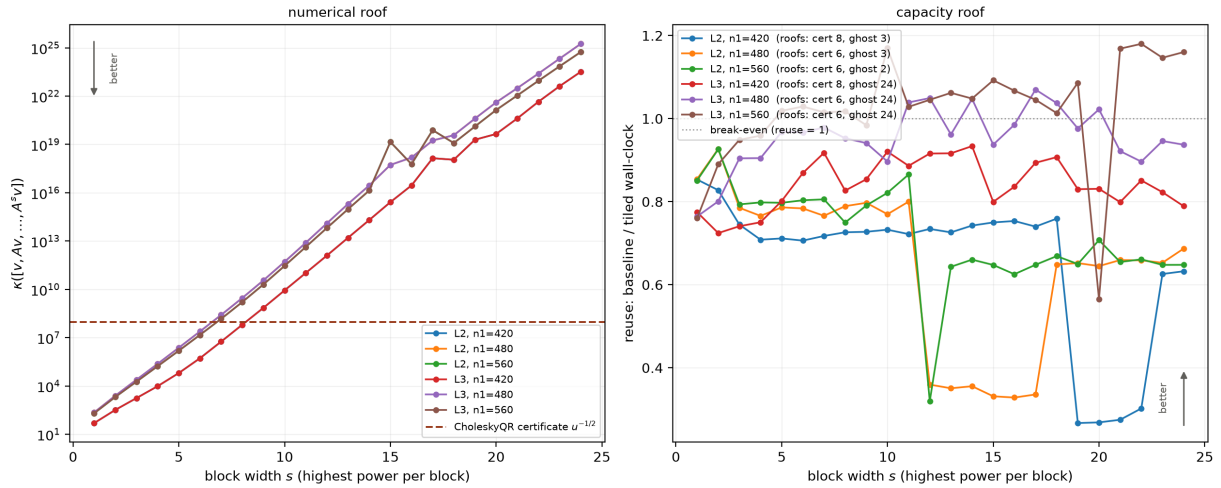


Figure 16: The two roofs on the block width s : the numerical certificate and the ghost-zone capacity, per tiling level.

The distinction is actionable. Where the certificate binds, a better-conditioned basis (Newton, Chebyshev [10, 5]) converts the halo's slack into real reductions. Where the ghost binds, only a thinner bandwidth or a coarser tiling level moves it—a better basis buys nothing. Pinning s assumes the first answer without measuring it.

8.11 THE MAP, AND WHERE THE REAL OPERATOR TRAVELS

The measured real-operator points anchor the map; a predicted continuation carries the operator to production grids. The continuation uses the *same* a-priori placement the instrument uses, and it is validated against the measured coordinates before it is trusted to extrapolate—the replica reproduces every measured (R_v, R_h) to better than 0.001% (Table 14, Figure 17).

Table 14: The real operator’s trajectory under grid refinement. Bisection on n puts the θ_v crossing at $n \approx 73$.

n	N	m	R_v	R_h	Source
10	1,000	6	0.0019	13.53	measured
20	8,000	8	0.0175	1.87	measured
25	15,625	8	0.036	0.904	predicted
61	226,981	12	0.582	0.080	predicted
74	405,224	13	1.063	0.047	predicted (first above θ_v)
120	1,728,000	15	4.74	0.0025	predicted

The interpretation of each leg:

- The measured points read $R_h > 1$ only because the grids small enough to dense-eigensolve ($n \leq 20$) sit far below any resolution a desk would price at. That is an irrelevant regime, not a grid where s -step pays.
- At production resolution the operator is squarely in **vertical-only** territory. The mechanism the application needs is matrix-powers tiling; the mechanism the certificate work supports is s -step. They are not the same corner.
- The upper-right corner is a **portability claim** on this hardware, not a measurement: reaching it needs an interconnect whose reduction cost is roughly $25\times$ puffin’s, which means crossing a device or node boundary rather than a CCX boundary.

8.12 STANDING CAVEATS

Carried explicitly so no figure is read as claiming more than it measures.

- DRAM bytes are **modeled, not counted** (puffin’s uncore counters need root); vertical-mechanism traffic claims are reconciled against the roofline, which is weaker evidence, and every such curve is labeled as a model.
- $s_{\max}$ is reported at a *chosen* tolerance of ± 1 step, not a measured cross-platform noise floor; block *counts* are quoted rather than raw $s_{\max}$. The cross-platform spread is one of the measurements the GPU study will supply (Section 9).
- $\kappa(X)$ is dense-eigensolvable only to $N \leq 32,768$: the certificate is validated well below where the performance claims live (real-operator control tops out at $N = 8,000$ against a production $N \approx 2.3 \times 10^5$). The structured computation lifts the cap for the synthetic operator only.
- The upper-right corner is unreachable on a single-NUMA socket; its status is a portability argument, not a measurement.

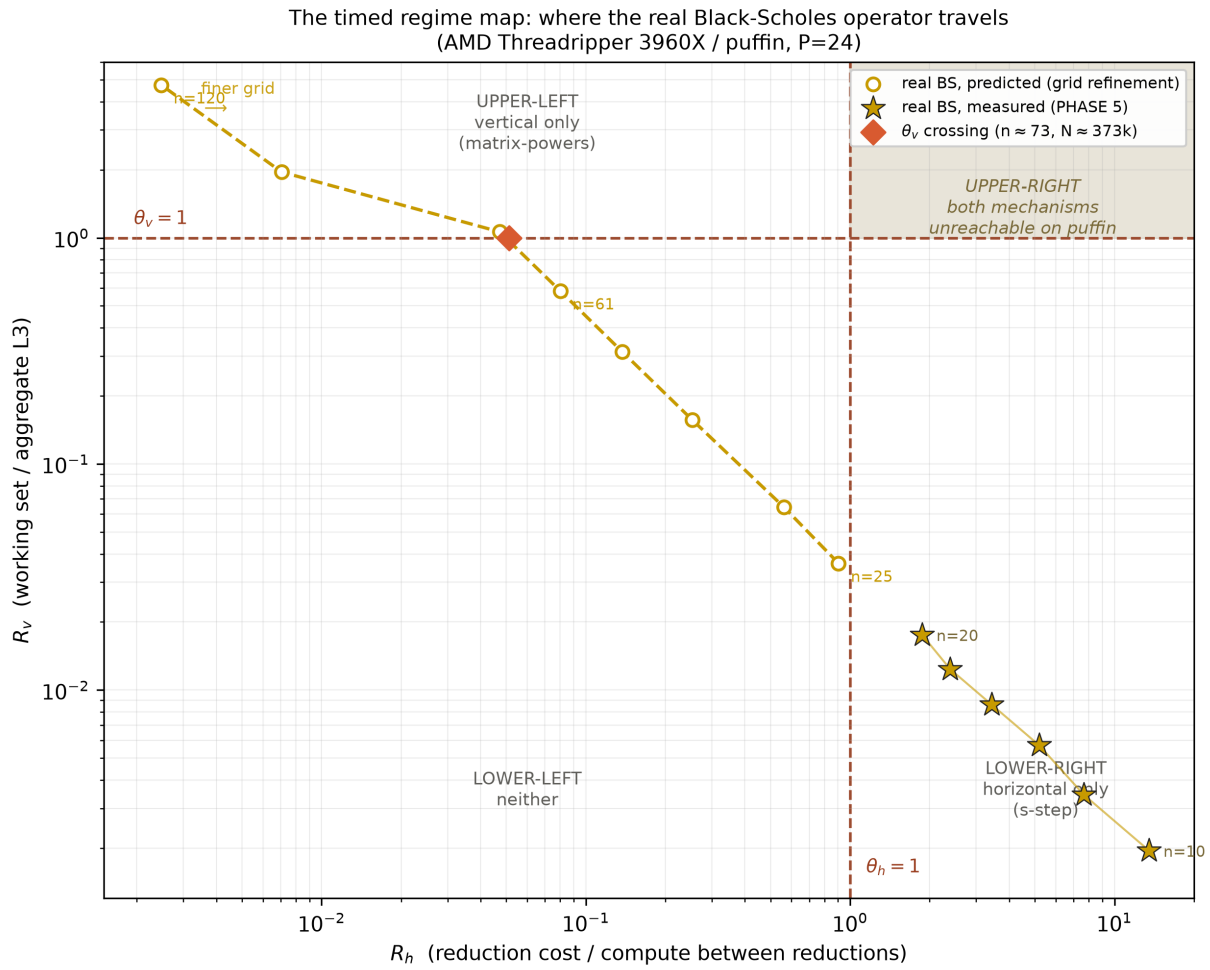


Figure 17: The regime map. The real Black-Scholes operator's refinement trajectory travels lower-right $\rightarrow$ lower-left $\rightarrow$ upper-left, crossing θ_v just under $n = 74$ and never entering the upper-right corner.

- Coordinates are **not portable across memory systems**: both axes must be re-derived from a new machine’s constants before any port. A GPU’s small L2, large bandwidth, and cheap on-chip reductions move both—which is exactly what Section 9 sets out to exploit.

9 TOWARD GPUS: FALSIFYING THE MAP

This section describes work in progress: the design is fixed and the first instruments exist, but no GPU crossover has yet been measured. It is included because the design is itself a result of the CPU study.

9.1 THE FALSIFICATION TEST

The CPU study’s central claim is that R_v and R_h are *dimensionless*—ratios, not times—so a reader can place their own operator and machine on the map. That claim has never been tested against a machine whose memory system differs in kind: puffin is one socket, one NUMA node, one cache hierarchy. The GPU work is therefore not a port. It is the falsification test of the map’s central claim, and both outcomes are informative: either the coordinates, recomputed from GPU constants, predict the measured GPU crossover at $\theta_v = \theta_h = 1$ and the dimensionless framing is vindicated; or they do not, the map is scoped to CPU-like memory systems, and *why* it fails names the missing term. The deliverable is a placed and *verified* crossover, not a running GPU kernel.

Two available GPUs make a near-controlled pair, and the pair is a better instrument than either card alone (Table 15): the memory systems are nearly identical—same 6 MB L2, same bandwidth class—while the FP64 rate differs by $\sim 12\times$. The two cards hold the map’s vertical axis roughly fixed and vary only the compute roof.

Table 15: The two-GPU controlled pair.

	L2	Memory BW	FP64 peak	FP64:FP32
RTX 3090 (consumer, puffin)	6 MB	~ 936 GB/s	~ 0.56 TFLOP/s	1/64
Tesla V100 (datacenter, cluster)	6 MB	~ 900 GB/s	~ 7.0 TFLOP/s	1/2

That lever tests the map’s hidden precondition. Both axes are *communication* axes, presupposing the kernel is on the memory side of the roofline; a hard-throttled FP64 rate does not move a point around the map, it throws the point *off* the map into compute-bound territory the map does not chart. Placement is therefore gated by an explicit roofline check—is $AI \cdot BW_{\text{achieved}} < \text{FLOP}/s_{\text{peak}}$ in the relevant precision?—evaluated before the map is read. The gate is itself a prediction the pair measures: **pass on the V100** (FP64 memory-bound), **fail on the 3090** (FP64 compute-bound). A map that predicts where it itself stops applying is a stronger vindication of the dimensionless framing than any single-machine crossover; the 3090 is the designed negative arm of the experiment, not a hazard to route around.

9.2 WHAT TRANSFERS, AND WHAT BREAKS

Everything machine-independent transfers unchanged: the synthetic operator, the tensor law (8.6.1), the numerics gates C1–C6, the reduction ledger, the CholeskyQR2 certificate, and the thresholds $\theta_v = \theta_h = 1$ themselves—a ratio crosses 1 where supply meets demand, which is what makes the thresholds portable. Only the coordinates are recomputed. Three things break structurally, not by constant swap:

- **The denominator of R_v .** On the CPU each engaged CCX brings its own L3 slice, so aggregate last-level cache scales with the team. A GPU’s L2 is a single fixed block shared by every streaming multiprocessor: engaging more of the device adds zero cache, so within one device R_v stops being a function of the team size at all. This is the event that *decouples the two axes*. On the CPU both coordinates shared the parallelism knob ($R_v \sim 1/P$, $R_h \sim P \log P$), which is why $R_v \cdot R_h$ collapsed to a P -independent constant (~ 0.04 on puffin)—the invariant that made the upper-right corner unreachable on one node. If the decoupling holds, that cancellation is broken and the corner may open for structural reasons rather than because of a slow interconnect; whether it does is the highest-impact open question, and it is being resolved on paper before any kernel is written.
- **The numerator of R_h .** The CPU model has two reduction tiers and one saturation constant (8.1.1). A GPU reduction crosses at least five qualitatively different tiers (Table 16), and the grid tier carries kernel-launch latency, which has no CPU analogue—a fixed cost per reduction independent of tree depth. Two parameters cannot express five tiers: the machine model needs a tier vector, and the reduction-cost predictor needs to walk it. The ladder is the instrument, not an obstacle: three reachable rungs with distinct costs (on-device, intra-node across PCIe, inter-node across the fabric) place the *same* problem at three different R_h values, so holding the problem and device fixed while changing only the reduction hardware traverses the horizontal axis directly, converting θ_h from a shaded prediction into a measured crossing.
- **The reuse model.** The CPU scatter knob costs a factor of a few in cache-line reuse; on a GPU the same permutation defeats warp coalescing and can serialize a 32-thread gather into 32 transactions. The scatter axis becomes *more* discriminating, but the reuse model must be recalibrated from scratch.

Table 16: The reduction-tier ladder on the target cluster (two nodes, one Tesla V100 per node, PCIe interconnect without NVLink). Every tier cost must be measured, never assumed.

Tier	Mechanism	Scope	Cost class
Warp	register shuffle	on-die	near-free, no memory traffic
Block	shared memory + barrier	on-die	cheap, on-chip
Grid	atomics or a second launch	on-die	expensive; carries launch latency
Device	PCIe + cross-socket link	intra-node	expensive; a genuine cross-socket hop
Node	network fabric	inter-node	dominant; fabric identity to confirm

9.3 PHASING

The study preserves predict-then-measure throughout: coordinates are predicted a priori from recomputed constants and never fitted to the measurement. Phase 0 (paper, no code) resolves whether the $R_v \cdot R_h$ invariant survives the chosen definition of team size, and whether an upper-right window even exists under the 16 GB device-memory ceiling at a resolution the pricing problem would actually use. Phase 1 builds the calibration instruments—a per-tier reduction calibration, a GPU STREAM for the achieved bandwidth roof, and an interconnect probe. Phase A lands the single-GPU vertical crossover and evaluates the roofline gate on both cards (predicted: V100 on-map, 3090 off-map); it also supplies the cross-platform $s_{\max}$ spread that Section 8.12 lists as missing, since a GPU is a maximally different floating-point platform. Phase B traverses the reduction ladder for the horizontal crossover and the

upper-right corner—the corner puffin could not reach, either reached-and-measured or shown empty with a mechanism. Phase C places the real Black–Scholes operator at production resolution on the V100 map and delivers the two-GPU falsification verdict.

9.4 OPEN QUESTIONS AND RISKS

Four items are recorded here because each would change the design, and each is a finding either way it resolves. First, the $R_v \cdot R_h$ invariant question above: if the decoupling holds, the upper-right corner opens structurally; if a sensible definition of team size re-couples the axes, the corner stays governed by the same invariant with a new constant. Second, whether an upper-right window exists at all under the 16 GB device-memory ceiling at usable resolution: on the CPU, a corner that “opened” turned out to be a 1%-wide artifact of freezing one axis while sliding the other, so the three constraints (spill the 6 MB L2 with margin, keep the reduction dominant, fit in 16 GB) must be solved jointly before a run is aimed at the intersection. If the window is empty, the honest claim is a *measured* boundary rather than a *reached* corner. Third, kernel-launch latency: if it dominates every grid-level reduction, the tier model collapses to a constant and the horizontal mechanism becomes uninteresting—itsself a result, but a different one. Fourth, occupancy: shared-memory tiling trades against it, and if that tradeoff proves independent of both coordinates, the map does not extend to GPUs without a third dimension—which would be stated, not hidden by fixing occupancy at one value.

10 CONCLUSION

This thesis set out to decide when communication-avoiding reformulations of a Krylov exponential integrator pay for a multi-asset option-pricing problem, and to make the answer portable. The instrument for that answer is a dimensionless regime map whose coordinates are predictors—computed from the problem, the spectrum, and machine constants before any run—and whose boundaries are tested by measurement.

On the numerical side, the certified s -step block width is predicted from the spectrum alone and verified integer-exact on 20 of 21 runs; a confound control shows safety is a property of the spectrum, not of the Krylov dimension; a log-robustness law bounds the damage non-normality does to the prediction (slope 0.069 in decades per decade of eigenvector conditioning for the constant-coefficient family, $4\times$ steeper for variable coefficients, the real operator under one s -step everywhere); and a closed-form tensor law gives the geometric growth of $\kappa(X)$ with asset dimension, along with the warning that the dense eigensolver value is meaningless once the spectrum degenerates.

On the measurement side, the study delivers three placed results. The horizontal mechanism is a null result with a mechanism: an $11\text{--}16\times$ cut in reduction count moves wall-clock by nothing, because the reduction primitive was never the bottleneck and Gram–Schmidt is bandwidth-bound—exactly as the map’s $R_h = 0.080$ predicted. The vertical mechanism is a shortfall with a mechanism: cache-blocked matrix powers loses to its own halo arithmetic at L2 tile sizes and to a residency-insensitive cap at L3, so the predicted crossover θ_v remains untested by this kernel rather than refuted. And the real operator’s trajectory under refinement—lower-right, lower-left, upper-left, crossing θ_v near $n = 74$ —says the mechanism the application needs at production resolution is matrix-powers tiling, not s -step, and that the upper-right corner is structurally unreachable on a single-NUMA socket.

10.1 FUTURE WORK

Three lines of work follow directly from the measured boundaries, beyond the GPU study already outlined.

A better-conditioned basis where the certificate binds. The block-width analysis (Section 8.10) shows the numerical roof binding at L3 tile sizes while the ghost roof still has slack. There, and only there, replacing the monomial basis with a Newton or Chebyshev basis [10, 5] converts that slack into real reductions; where the ghost binds, a better basis buys nothing. The map now says where each investment pays.

Counted, not modeled, DRAM traffic. The vertical-mechanism evidence is reconciled against the roofline because puffin’s uncore counters require root access. Repeating the sweep on a machine with accessible memory controllers would upgrade the traffic claims from model to measurement—and either close or widen the gap between the 4–8× modeled ceiling and the measured shortfall.

A structural result for the real-operator certificate gap. The tensor identity closes the computability gap for the synthetic operator because that operator is a pure Kronecker sum; the real Black–Scholes operator with all-pairs correlation is not. Brute force does not help—at production resolution the eigenvector matrix alone would need hundreds of gigabytes—so what is needed is a decomposition of the correlated operator as a Kronecker backbone plus a correlation perturbation, bounding its $\kappa(X)$ as a perturbation of the factorizable part. That is an analysis problem, not a bigger eigensolve.

The map’s central claim—that its coordinates transfer—remains its least-tested one, and the GPU study of Section 9 is designed to falsify it on a memory system that differs in kind, with a controlled two-GPU pair whose roofline gate the map must predict. That work is in progress; the map now exists to tell it what to build.

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